Ruikai He (Carry)

Camera Imaging and ISP Algorithm Researcher | Color Science | Computational Photography
hurricane9737@gmail.com | (+852) 9572 1307 | Hong Kong | Open to relocation
Portfolio | Google Scholar | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Camera imaging and color science researcher with 4+ years of experience developing learning-based and model-based algorithms for camera image signal processing (ISP), automatic white balance, tone mapping, RAW/DNG color calibration, and multi-camera color consistency. Builds reproducible prototypes in Python, MATLAB, PyTorch, and OpenCV, with experience evaluating image quality using PSNR, SSIM, LPIPS, CIELAB color difference, and angular error. Ph.D. oral defense completed; degree expected November 2026.

TECHNICAL SKILLS

Programming: Python, MATLAB
Camera ISP and Color Science: RAW/DNG workflows, profile gain table maps (PGTM), automatic white balance (AWB), computational color constancy (CC), tone mapping (TM), color correction matrices (CCMs), multi-camera color consistency, camera calibration, sensor spectral response.
Platforms: iOS Metal (exploratory)

WORK EXPERIENCE

Research Associate Sep 2025 - Present
The Hong Kong Polytechnic University | Color, Imaging, and Illumination Laboratory

- Develop and evaluate camera-side image-processing algorithms for tone mapping, color constancy, multi-camera color consistency, and perceptual image-quality analysis using Python, MATLAB, PyTorch, and OpenCV.
- Collaborate with imaging-industry partners on algorithm development, technical demonstrations, and preliminary deployment feasibility, including exploratory use of Metal for on-device computational optimization.
- Communicate research outcomes through reproducible prototypes, technical presentations, and peer-reviewed publications.

SELECTED RESEARCH PROJECTS

Image-Adaptive Profile Gain Table Map Generation Sep 2025 - Jun 2026

- Designed a U-Net-based tone-mapping model that separates global prediction from local adjustment and converts the predicted local curves into Profile Gain Table Maps compatible with Adobe's DNG workflow.
- Implemented the training and evaluation pipeline in PyTorch using RMSE, spatial and depth total-variation losses, and a custom shape constraint; assessed output quality using gain-ratio distributions, PSNR, SSIM, LPIPS, CIELAB color difference, and artifact inspection.
- Collaborated with Leica to define algorithm requirements and evaluated deployment feasibility on iOS, including exploratory Metal optimization to reduce computational cost.

Integration of RGB Images and Ambient Light Sensor Data for Color Constancy Mar 2025 - Oct 2025

- Extended C5, PCC, and U-Net-based color-constancy models to integrate multispectral ambient-light-sensor data through latent-feature projection, feature addition, and concatenation while preserving the original RGB branches.
- Evaluated the models on more than 8,000 paired RAW images and 13-channel ALS measurements using RGB-only, ALS-only, dual-input, and modality-masking experiments.
- Demonstrated that incorporating ALS data generally reduced illuminant-estimation angular error, with the largest improvements occurring in the worst 5% of challenging illumination cases.
- Developed a proof-of-concept implementation and public project demonstration to communicate the method and its behavior.
- Authored the research paper and delivered an oral presentation at the 33rd Color and Imaging Conference (CIC33).
- Open-Source Implementation: https://github.com/Hurricane-k/Demo-ALS4IllumEstimation

Color Consistency for Multi-Camera Systems Sep 2023 - Sep 2024

- **White-Point Alignment:** Developed a real-time RAW-space conversion method that aligns camera white points using a limited set of carefully selected calibration illuminants.
- Evaluated the method on 756 ordered camera pairs constructed from 28 public camera-sensitivity datasets and 544 representative illuminants, then verified improved cross-camera color consistency using DNG images.
- Analyzed performance across camera pairs and found that alignment accuracy depends on the linear relationship between their color responses, providing a practical indicator of method suitability.
- **Open-Source Implementation**: https://github.com/Hurricane-k/Demo-AlignWhitePoints
- **Image-Adaptive RAW-to-RAW Mapping:** Estimated scene-specific color transformation matrices from unregistered main- and telephoto-camera images using spatial or color-space sampling and least-squares fitting; evaluated cross-camera consistency using ColorChecker CIE 1976 color difference.
- **Joint CCM Calibration:** Jointly optimized the color correction matrices of a camera pair to balance absolute colorimetric accuracy against cross-camera consistency, producing fixed matrices suitable for deployment in camera calibration pipelines.

RAW-to-RAW Mapping for Cross-Camera Color Constancy Sep 2023 - Sep 2024

- Developed an offline RAW-to-RAW mapping framework that transforms reference-camera images and illuminant labels into target-camera training data.
- Retrained lightweight, camera-specific color-constancy models using the synthesized data and evaluated them on real target-camera images using angular error, reducing the need for target-camera data collection.

EDUCATION

The Hong Kong Polytechnic University Sep 2022 – Nov 2026 (expected)
Ph.D. Candidate in Color Imaging, GPA: 3.61/4.30

- Advisor: Prof. Minchen Wei
- Dissertation: Color Correction for Advanced Imaging Systems

Tongji University Sep 2019 - Mar 2022
Master of Engineering in Heating, Ventilation and Air Conditioning, GPA: 4.68/5.00

Army Engineering University of PLA Sep 2015 - Jun 2019
Bachelor of Engineering, Overall Average: 90.2/100

SELECTED PUBLICATIONS

Wei, M., **He, R.**, & Yue, J. (2026). Beyond angular error: Rethinking metric for white balance and its effect on image color appearance. Color Research & Application, 51(3).

He, R., & Wei, M. (2025). Integration of RGB image and ALS for color constancy. Color and Imaging Conference, 33(1), 80-85.

He, R., Wei, M., Qu, X., & Hu, T. (2025). Color correction for multi-camera systems: Aligning white points. Journal of the Optical Society of America A, 42, 690-698.

AWARDS

Outstanding Graduate, Tongji University Apr 2022
Outstanding Student of Tongji University, Academic Year 2021-2022 Oct 2021
National Fellowship for Postgraduate Students Oct 2021